PROJECT REPORT

AWS Multi-Region Warm Standby

Disaster Recovery Architecture

High availability, data resilience and cost optimization across Mumbai (ap-south-1) and Singapore (ap-southeast-1)

Item	Detail
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AWS Account	343644157719
Primary Region	Asia Pacific (Mumbai) - ap-south-1
Secondary Region	Asia Pacific (Singapore) - ap-southeast-1
Date	October 24, 2026
Services	EC2, S3, IAM, VPC, Route 53

Table of Contents

1. Abstract 3
2. Objectives 3
3. Architecture Overview 3
4. AWS Services and Implementation 4
4.1 Amazon EC2 - Compute Tier 4
4.2 Amazon S3 - Data Tier 5
4.3 AWS IAM - Security 7
4.4 Amazon VPC - Networking 7
4.5 Amazon Route 53 - DNS and Health Monitoring 9
5. Validation and Test Plan 10
6. Results and Observations 12
Items to review 12
7. RPO and RTO 12
8. Cost Optimization 12
9. Challenges and Lessons Learned 13
10. Future Enhancements 13
11. Conclusion 13

1. Abstract

This project implements a **Warm Standby Disaster Recovery (DR)** architecture on AWS across two geographic regions. Mumbai (`ap-south-1`) is the Primary Active Region and Singapore (`ap-southeast-1`) is the Secondary Backup Region. A web application runs on a standalone EC2 server in Mumbai and receives 100% of live traffic during normal operation. In the background, S3 Cross-Region Replication keeps the data tier synchronized with Singapore, where a pre-provisioned standby server waits to take over if Mumbai fails.

The design provides a low Recovery Point Objective (RPO) and a low Recovery Time Objective (RTO) without paying for a fully provisioned, duplicate compute tier. This report documents the architecture, the implementation of each AWS service with console evidence, the validation procedure, observations, and recommendations for improvement.

2. Objectives

- Build a highly available web hosting setup that survives a complete regional outage.
- Keep data continuously synchronized between regions with no manual intervention.
- Secure server-to-storage access with IAM roles instead of hard-coded credentials.
- Automate failover decisions using Route 53 health checks and a failover routing policy.
- Minimise cost by using free-tier eligible instances and an idle standby environment.
- Validate the design with a simulated regional outage.

3. Architecture Overview

During normal operations, users reach the primary web server in Mumbai. The server reads its content from the primary S3 bucket. S3 Cross-Region Replication copies every new or changed object to a bucket in Singapore. The two VPCs are joined by a VPC peering connection, and Route 53 health checks continuously probe both web endpoints. If Mumbai becomes unavailable, the health check fails and the failover routing policy shifts traffic to the Singapore warm standby, which serves the replicated data.

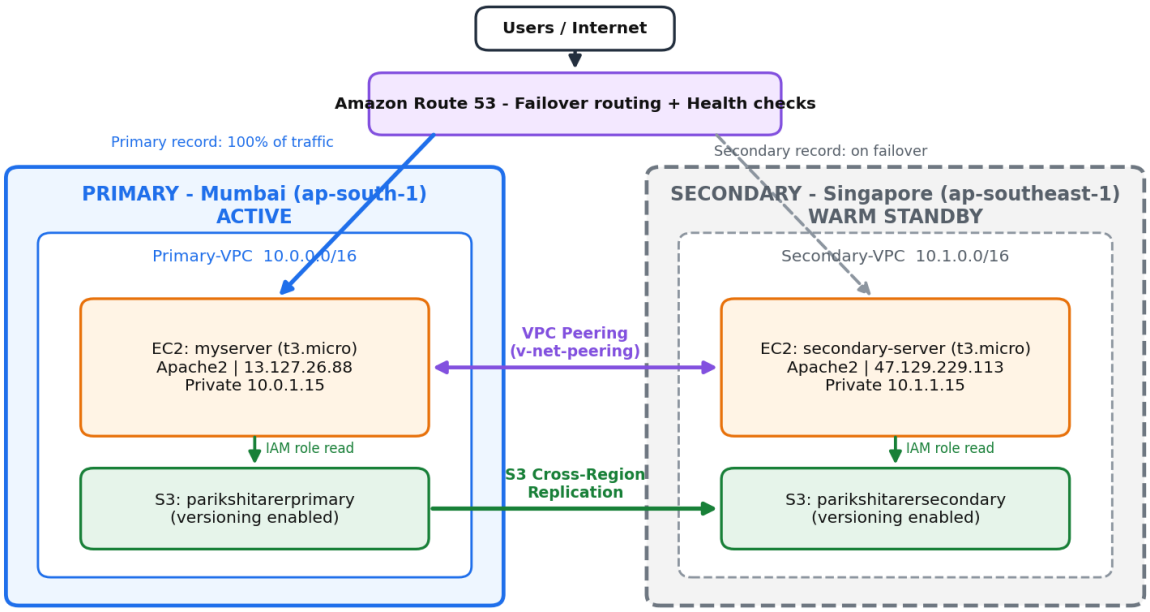


Figure 1: *High-level warm standby DR architecture*

Component	Primary (Mumbai)	Secondary (Singapore)
Region	ap-south-1	ap-southeast-1
Role	Active	Warm Standby (passive)
EC2 instance	myserver (i-08259ebb457521131)	secondary-server (i-069c9bc4ee0fb6b68)
Instance type	t3.micro	t3.micro
Public IPv4	13.127.26.88	47.129.229.113
Private IPv4	10.0.1.15	10.1.1.15
VPC	Primary-VPC (vpc-06d19b3622a194305)	Secondary-VPC (vpc-0fd0d2bdf53298295)
VPC CIDR	10.0.0.0/16	10.1.0.0/16
S3 bucket	parikshitarerprimary	parikshitarersecondary
IAM role	EC2-S3-Read-Role	EC2-S3-Read-Role

4. AWS Services and Implementation

4.1 Amazon EC2 - Compute Tier

Two standalone Ubuntu instances host the application, one per region, both on low-cost `t3.micro` shapes. Each instance is launched with a **User Data script** that runs at first boot to update packages, install the Apache2 web server and the AWS CLI, pull data objects from S3, and generate the frontend page dynamically.

- The Mumbai server (myserver) runs in a public subnet of Primary-VPC and serves the primary page.
- The Singapore server (secondary-server) runs in a public subnet of Secondary-VPC as the warm standby.
- Both expose HTTP port 80 through Security Groups, and IMDSv2 is set to Required for hardened metadata access.

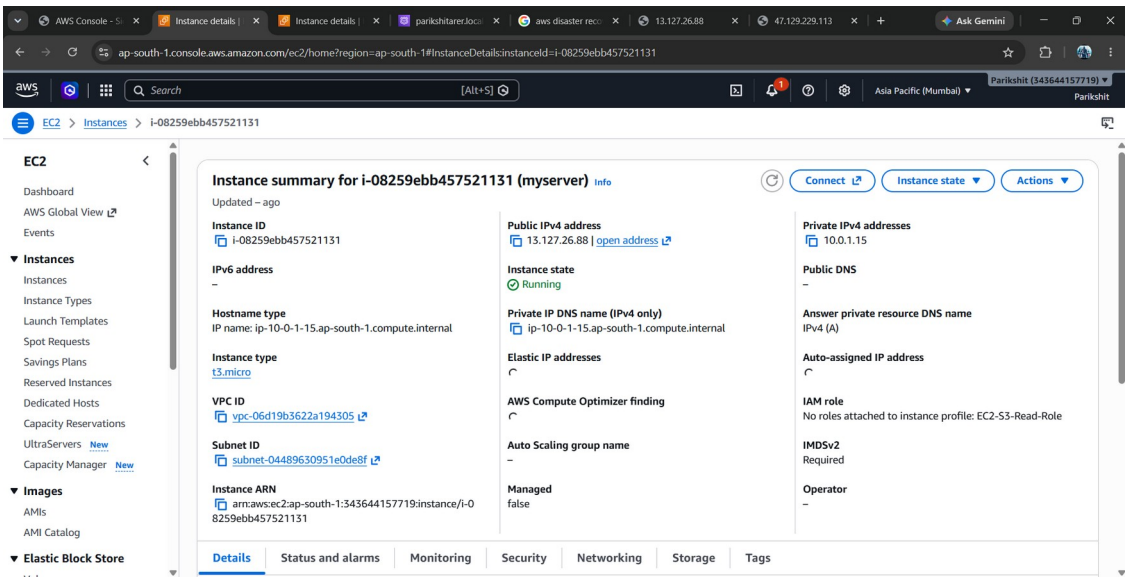


Figure 2: *Primary EC2 instance (myserver) in Mumbai, public IP 13.127.26.88*

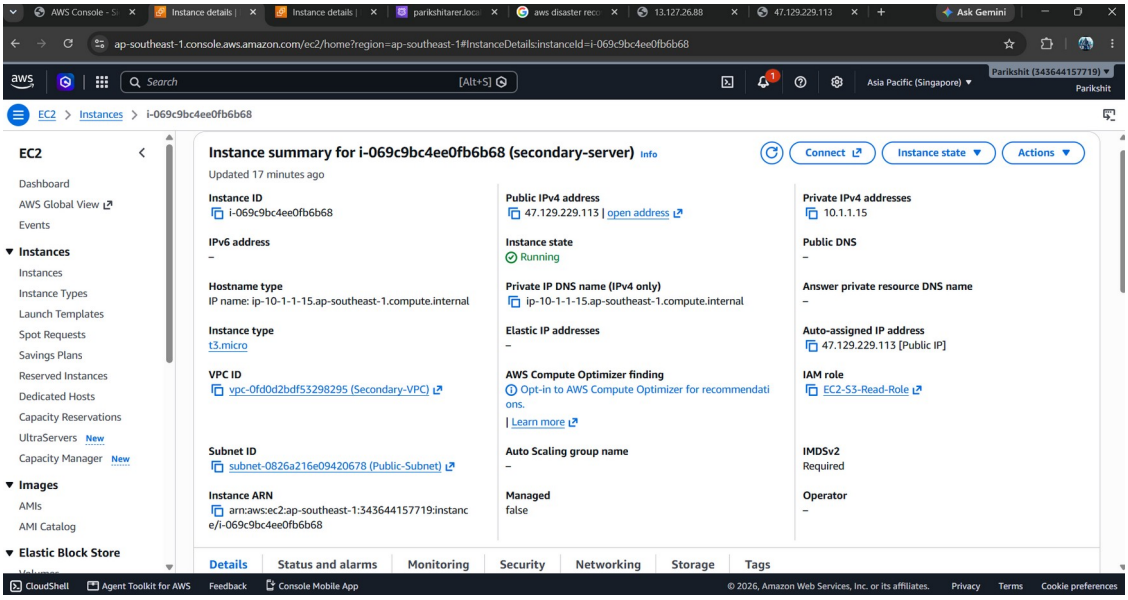


Figure 3: *Secondary EC2 instance (secondary-server) in Singapore, public IP 47.129.229.113*

4.2 Amazon S3 - Data Tier

Two versioning-enabled buckets form the durable data layer: `parikshitarerprimary` in Mumbai and `parikshitarersecondary` in Singapore. Both were created on October 23, 2026.

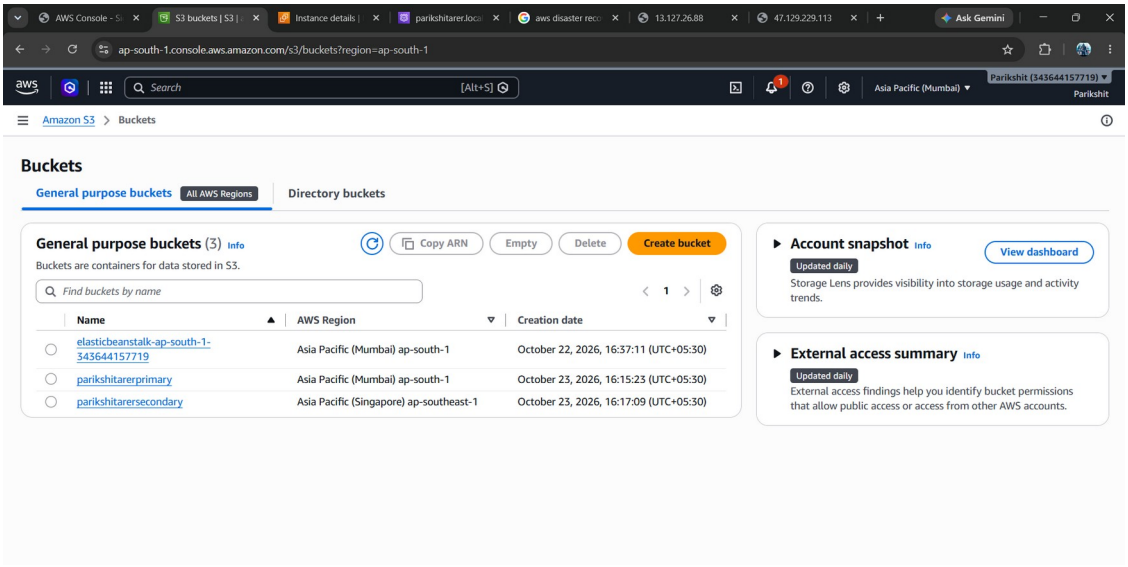


Figure 4: *General purpose buckets: elasticbeanstalk (default), parikshitarerprimary and parikshitarersecondary*

A replication rule named **cross-region** is configured on the primary bucket. Every object created or modified in the primary bucket is copied asynchronously over the AWS backbone to the secondary bucket.

Setting	Value
Rule name / Status	cross-region / Enabled
Destination bucket	s3://parikshitarersecondary
Destination Region	Asia Pacific (Singapore) ap-southeast-1
Scope / Priority	Entire bucket / 0
Storage class / Replica owner	Same as source / Same as source
Replication Time Control	Disabled
Replica modification sync	Disabled

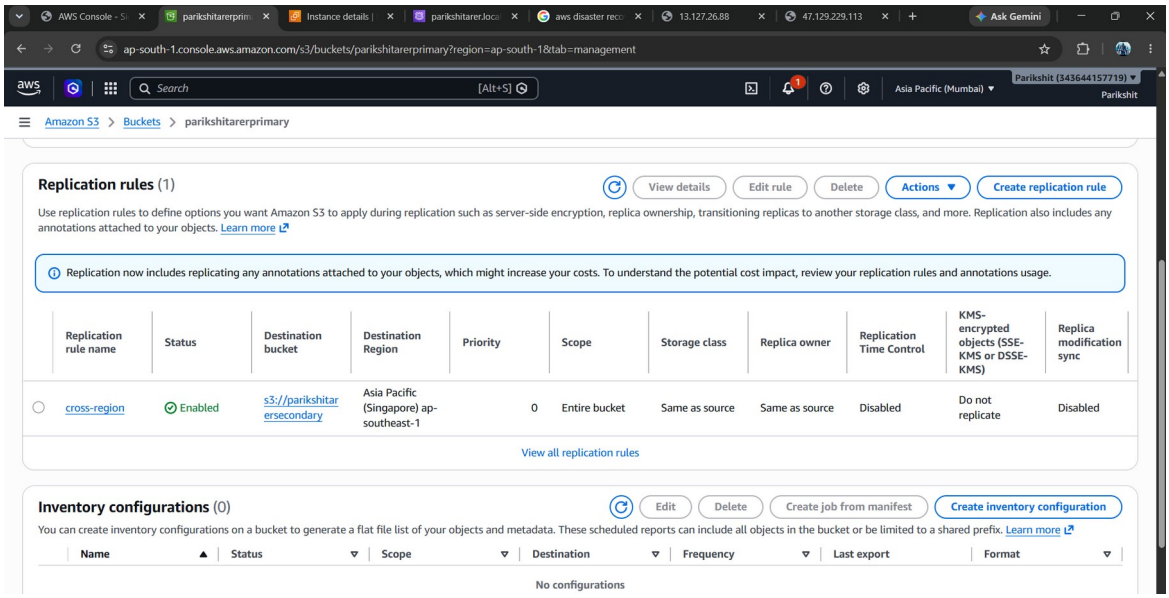


Figure 5: *S3 replication rule "cross-region" on the primary bucket*

4.3 AWS IAM - Security

A custom IAM role and instance profile, **EC2-S3-Read-Role**, is attached to the EC2 instances with the managed policy `AmazonS3ReadOnlyAccess`. This follows the principle of least privilege: the servers read from S3 through the AWS CLI using temporary credentials delivered by the instance metadata service, so no access keys are stored in code or on disk.

4.4 Amazon VPC - Networking

Two isolated networks were built with an identical layout: four subnets across two Availability Zones (a public and a private subnet in each), two route tables, and an Internet Gateway for public connectivity. Security Groups act as stateful host-level firewalls that allow HTTP port 80 from the internet.

Attribute	Primary-VPC (Mumbai)	Secondary-VPC (Singapore)
CIDR	10.0.0.0/16	10.1.0.0/16
Availability Zones	ap-south-1a, ap-south-1b	ap-southeast-1a, ap-southeast-1b
Subnets (4)	Public-Subnet, Private-Subnet, Public-Subnet-2, Private-subnet-2	Public-Subnet, Private-subnet, Public-subnet-2, Private-subnet-2
Route tables (2)	Route-1, rtb-00cbe3acdc0a20f22	Route-2, rtb-0d2ba96021f728e8b
Internet Gateway	IGW-1	IGW

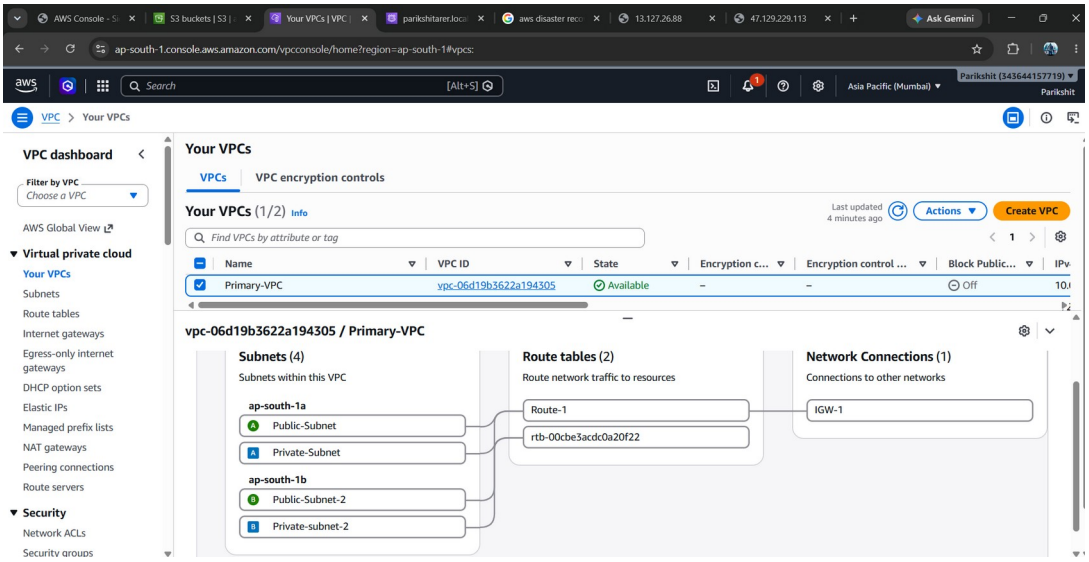


Figure 6: *Primary-VPC resource map (Mumbai)*

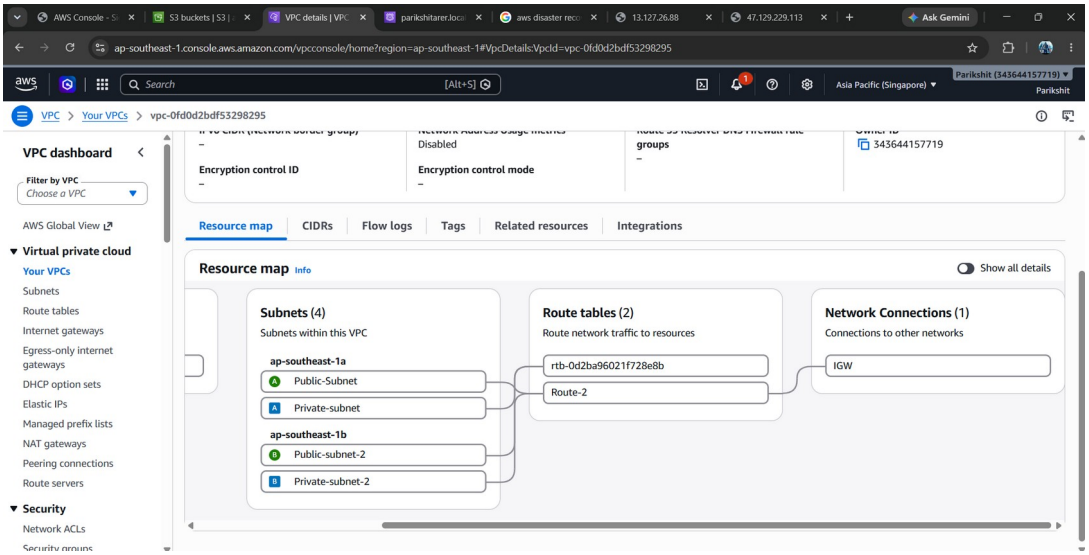


Figure 7: *Secondary-VPC resource map (Singapore)*

VPC Peering. A cross-region peering connection named `v-net-peering` (`pcx-01d29206872b1379e`) was requested from Primary-VPC (requester, 10.0.0.0/16) to Secondary-VPC in Singapore, then accepted from the Singapore console. This gives the two networks a private communication path over the AWS backbone.

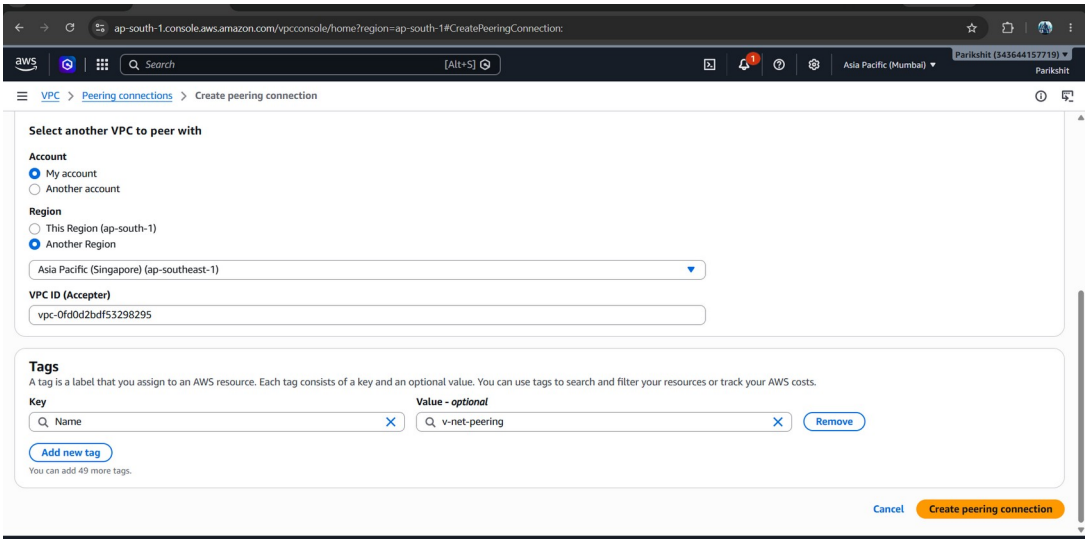


Figure 8: *Creating the peering connection from Mumbai to the Singapore VPC*

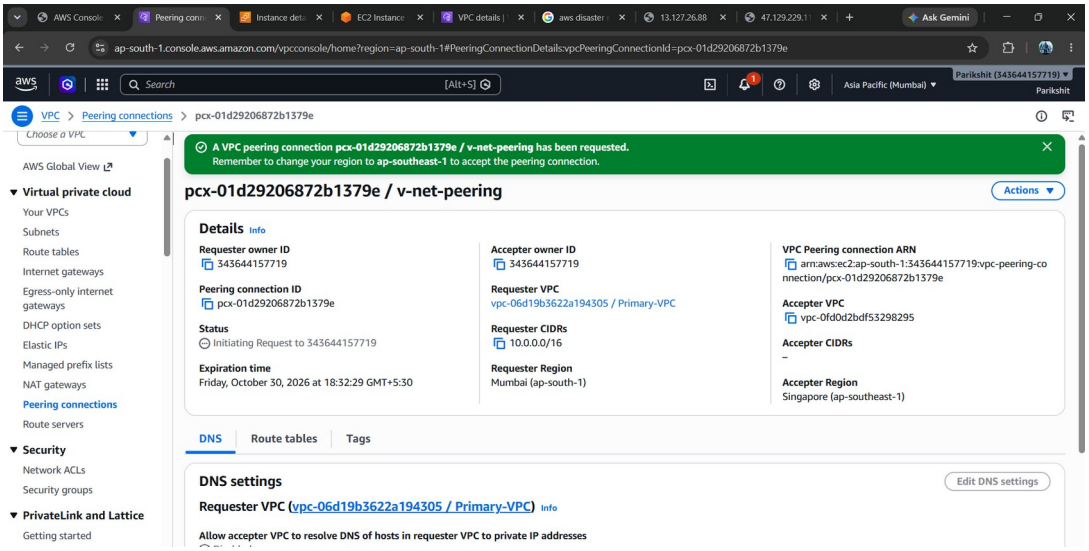


Figure 9: *Peering connection requested; status "Initiating Request"*

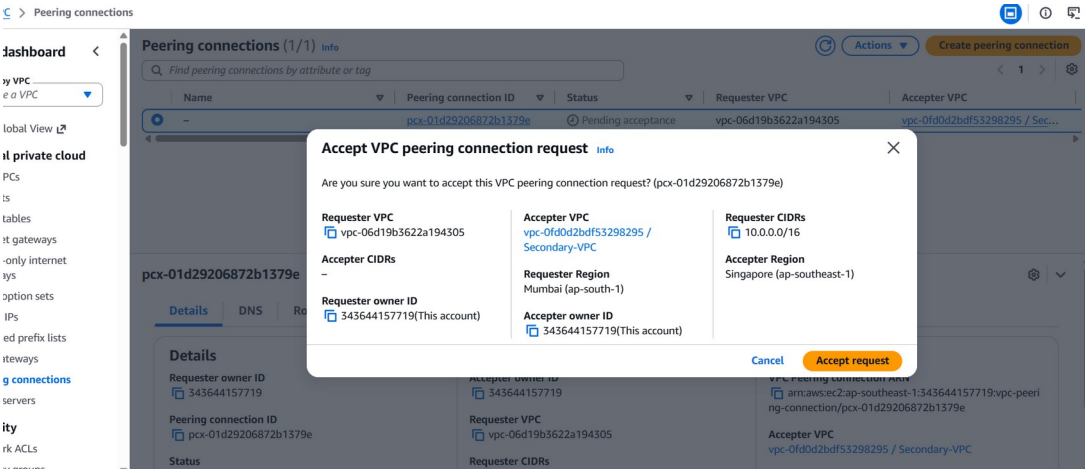


Figure 10: *Accepting the peering request in the Singapore region*

4.5 Amazon Route 53 - DNS and Health Monitoring

A **Private Hosted Zone** named `parikshitarer.local` holds four records: the default NS and SOA records, and two A records for `app.parikshitarer.local` using the **Failover** routing policy. The Primary record points to `13.127.26.88` and the Secondary record to `47.129.229.113`, each with a 300-second TTL and an associated health check.

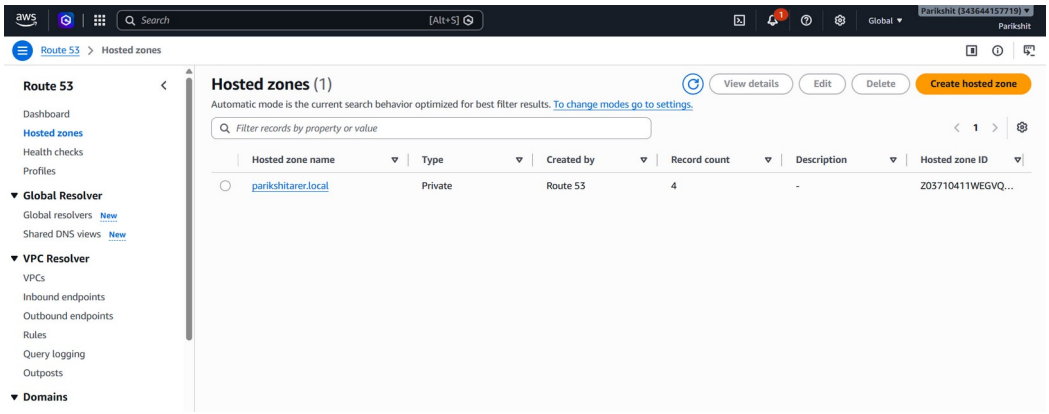


Figure 11: *Route 53 hosted zone parikshitarer.local (private)*

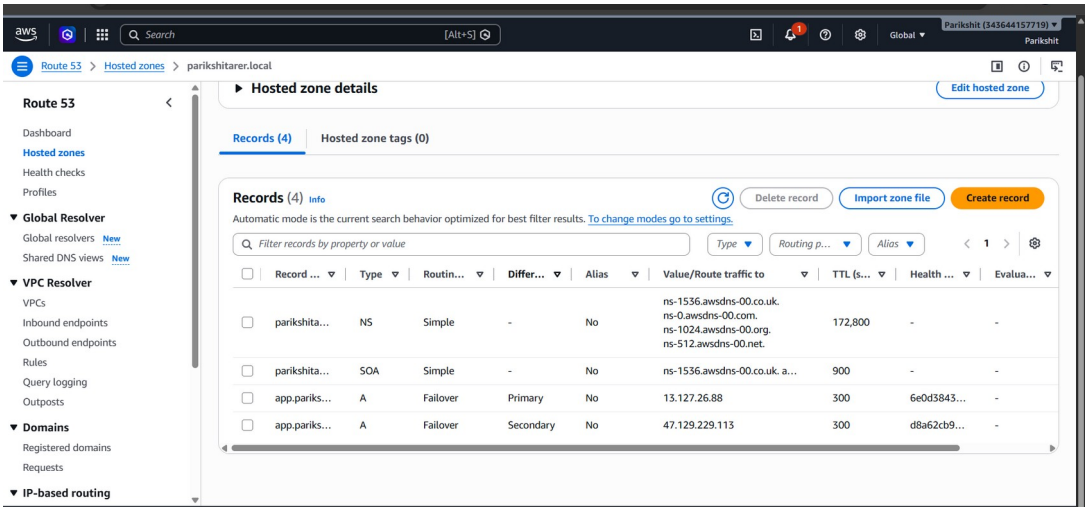


Figure 12: *Failover A records (Primary and Secondary) with health checks*

Two health checks are enabled, `Mumbai-server` (http://13.127.26.88) and `Singpore-server` (http://47.129.229.113). Both report **Healthy**.

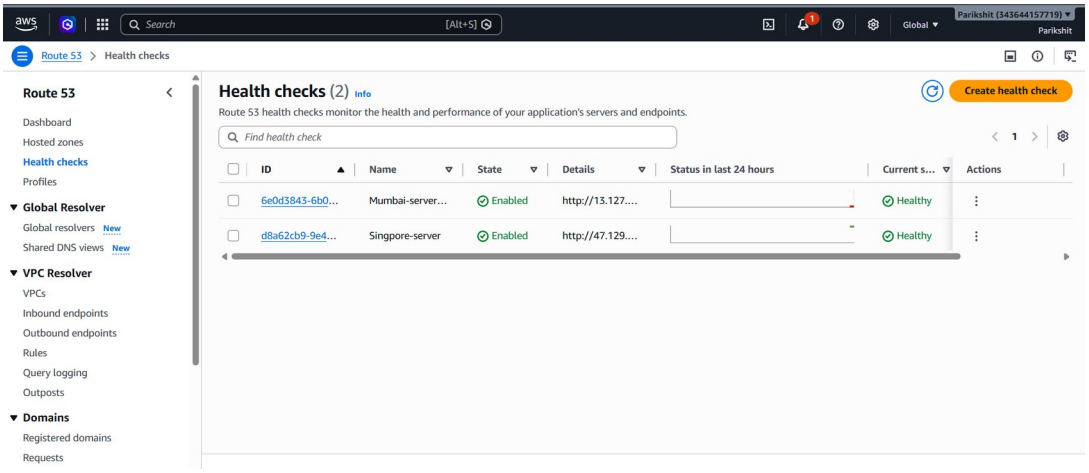


Figure 13: *Route 53 health checks for both regional endpoints*

5. Validation and Test Plan

#	Test	Method	Expected / Observed
1	Active routing	Open http://13.127.26.88; curl from the Singapore instance	Page shows "Hello from Primary Region (Mumbai)". Confirmed.
2	Standby reachable	Open http://47.129.229.113	Page shows "Hello from Secondary Region (Warm Standby - Singapore)". Confirmed.
3	Health monitoring	Check Route 53 health checks	Both endpoints Healthy. Confirmed.
4	Replication	Upload data.json to the primary bucket; verify in the secondary bucket	Object appears in Singapore.
5	Outage simulation	Stop the Mumbai EC2 instance	Mumbai health check turns Unhealthy; failover record resolves to Singapore.

#	Test	Method	Expected / Observed
6	Standby activation	Refresh the Singapore endpoint	Standby serves the synced data intact.

Evidence captured during validation is shown below. The curl test was run from an SSH session on the Singapore instance (private IP 10.1.1.15) against the Mumbai public IP.

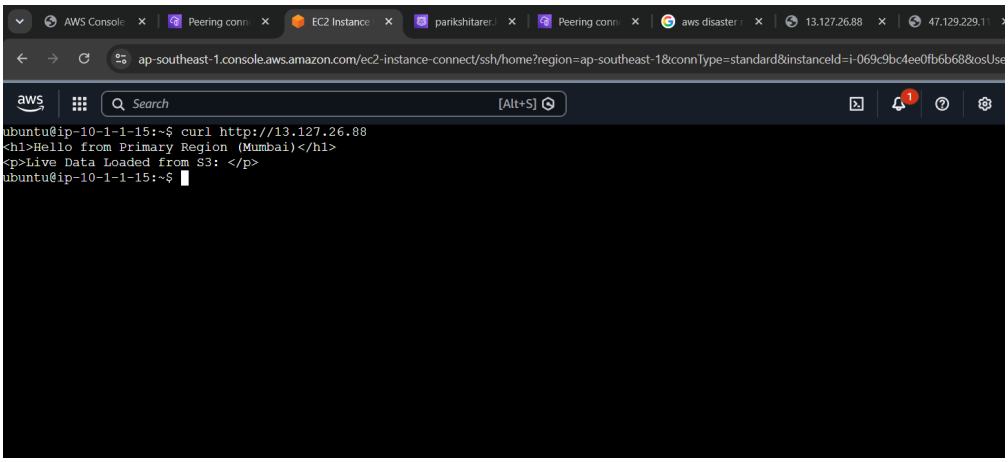


Figure 14: *curl http://13.127.26.88 returns the primary page*

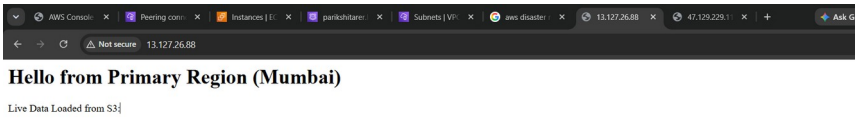


Figure 15: *Primary endpoint in the browser (Mumbai)*

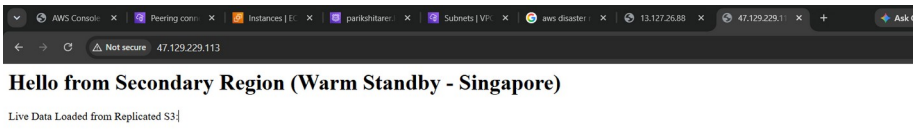


Figure 16: *Secondary endpoint in the browser (Singapore warm standby)*

6. Results and Observations

- Both regional web servers are running and reachable over HTTP.
- Route 53 health checks are active and Healthy for both endpoints.
- VPC peering was requested and accepted between Mumbai and Singapore.
- The S3 replication rule is enabled for the entire primary bucket.

Items to review

1. Empty S3 content. Both pages show "Live Data Loaded from S3:" with no value. On the Mumbai instance, the EC2 console reports that no roles are attached to the instance profile EC2-S3-Read-Role. The role should have AmazonS3ReadOnlyAccess attached and be confirmed on the instance; data.json must also exist in the bucket.

2. Existing objects. S3 replication copies only objects uploaded after the rule is enabled. Older objects need S3 Batch Replication.

3. Private hosted zone. A private zone resolves only inside the associated VPCs. External users still reach the sites by public IP. True public failover needs a public hosted zone with a registered domain.

4. Peering routes. After peering, route table entries (10.1.0.0/16 via the peering connection in Mumbai and 10.0.0.0/16 in Singapore) are required for private traffic to flow.

7. RPO and RTO

Metric	Value	Notes
RPO	Seconds to minutes	S3 CRR is asynchronous. Replication Time Control would add a 15-minute SLA.
RTO	Minutes	Determined by health check interval, failure threshold and the 300 s DNS TTL.
Standby cost	Low	One small instance in Singapore instead of a full duplicate environment.

8. Cost Optimization

- Free-tier eligible t3.micro instances in both regions.
- Warm standby keeps the secondary minimal instead of a full active-active deployment.
- No NAT gateways, load balancers or paid database tier in this design.
- Costs to monitor: S3 storage in both buckets, inter-region transfer for replication and peering, health checks, and public IPv4 addresses.

9. Challenges and Lessons Learned

- Cross-region peering must be accepted from the accepter region, and the request expires after seven days (October 30, 2026 here).
- IAM is global, but instance profile attachment is per instance and must be verified in each region.
- Console region selection is easy to get wrong across Mumbai, Singapore and global services such as Route 53.
- Versioning must be enabled on both buckets before a replication rule can be created.

10. Future Enhancements

- Add an Application Load Balancer and Auto Scaling group in Singapore so the standby scales out on failover.
- Use a public hosted zone with a custom domain for real public failover.
- Enable S3 Replication Time Control and CloudWatch alarms on replication metrics.
- Add cross-region database replication (Aurora Global Database or RDS read replica) for stateful workloads.
- Define the stack as code with CloudFormation or Terraform and run regular DR drills.
- Send SNS notifications on health check state changes.

11. Conclusion

The project demonstrates a practical, cost-conscious multi-region disaster recovery design on AWS. By combining EC2, S3 Cross-Region Replication, IAM roles, VPC peering and Route 53 failover with health checks, the architecture keeps data synchronized, secures access without static credentials, and can redirect users to a standby region within minutes of a regional failure. Once the IAM role attachment and S3 content items are resolved, the setup serves as a complete reference implementation of warm standby disaster recovery.